# Saksham Bansal

Atlanta, Georgia | 404-936-0025 | sakshambansal@gatech.edu | linkedin.com/in/bansalsaksham | bansalsaksham.github.io

## OBJECTIVE

Electrical Engineering senior at Georgia Tech focused on embedded systems and robotics, with experience in firmware, sensor integration, feedback control, and hardware/software debugging.

## EDUCATION

**Georgia Institute of Technology | Atlanta, GA** *August 2023 – Present*
Bachelor of Science in Electrical Engineering Expected Graduation, May 2027

## SKILLS

**Embedded Systems:** C/C++, Python, ARM Cortex-M (STM32), ESP32, Arduino, Linux, I2C, SPI, UART, PWM, TCP/IP
**Robotics & Controls:** Sensor Integration, Feedback Control, FSM Design, Motor/Actuator Modeling, IMU-Based Control
**Hardware & Debug:** PCB Design (KiCad, Altium), Oscilloscope, Logic Analyzer, DMM, Bench Power Supply, Hardware Bring-Up
**Tools:** MATLAB/Simulink, Git/GitHub, Wireshark, NI LabVIEW, Autodesk Inventor

## EXPERIENCE

**Promaxo Inc. | Oakland, CA** *June – August 2025*
***Robotics Engineering Intern***
*A MedTech company developing MRI-compatible robotic guidance systems for in-office prostate biopsies and treatment.*

- Developed MATLAB/Python models of pneumatic actuator dynamics, motor kinematics, and gear-train behavior for an MRI-compatible robotic positioning system.
- Ran 500+ parameter sweeps across pressure, hose length, gear ratio, and eccentricity, identifying configurations improving torque-speed-power tradeoffs by an estimated 10–15%.

**Laboratory for Intelligent Decision and Autonomous Robots (LIDAR) | Atlanta, GA** *May - June 2025*
**Undergraduate Research Assistant**

- Integrated high-torque PWM servos with a Linux-based robot controller (Intel NUC) to actuate a robotic arm.
- Diagnosed and resolved a UART vs. USB HID protocol compatibility issue blocking hardware integration; implemented Python-based USB HID control to simplify the system's wiring and control stack.

**Electronic ARTrium (Vertically Integrated Project) | Atlanta, GA** *January 2025 – May 2026*
***Undergraduate Research Assistant | Prof. Timothy Brothers***

- Engineered the core depth-sensing pipeline for an interactive AR sandbox; interfaced with an Orbbec depth camera and reconstructed 16-bit depth values via a custom bit-shifting algorithm, relayed to Unity for real-time terrain mapping.
- Built exhibit components and Arduino sensor nodes, debugging a JSON packet mismatch in the Arduino-Unity server pipeline.

## PROJECTS

**Autonomous Rescue & Inspection System (A.R.I.S.) | Innovate ECE 2026** *March 2026*
*Winner, $1,000 Best in Thread: An ESP32-S3 autonomous differential-drive rover for disaster-response hazard assessment.*

- Developed a non-blocking FSM integrating MPU6050 IMU, HC-SR04 ultrasonic, MQ-6 gas sensing, and L298N motor control for obstacle avoidance, hazard detection, and proportional heading correction.
- Implemented a live WebSocket telemetry dashboard streaming IMU heading, sonar clearance, and gas readings in real time.

**Rootry (formerly SoilHawk) | Automatic Irrigation System** *2019 - Present*
*UPenn EcoVenture Runner-Up (Worldwide): Solar-powered precision irrigation for smallholder farms, field-tested in three phases.*

- Designed an ESP32-S3 PCB (KiCad) with hybrid GSM/Wi-Fi connectivity and a real-time local dashboard via ESPAsyncWebServer.
- Programmed a non-blocking, sensor-driven pump control firmware (hysteresis-band moisture thresholds, adaptive temperature compensation) for off-grid solar-powered deployment.

**IEEE Robotics Team (VEX U)|Georgia Tech** *September 2023 - May 2025*

- Wrote C/C++ firmware for autonomous navigation and driver-controlled functionality, integrating sensor feedback (encoders, IMU) for closed-loop precision movement.
- Modeled custom chassis and manipulator mechanisms in Autodesk Inventor, iterating designs against on-field performance data.

## RELEVANT COURSEWORK

**Embedded Systems:** Hardware/Software Co-Design, MIPS Architecture, Real-Time Systems, Parallel Programming, Nonblocking I/O
**Signals & Systems:** Fourier & Laplace Transforms, Frequency Response, Filtering, Linear Systems, Z-Transform
**Control Systems:** Feedback, Stability Analysis, Root Locus, Bode/Nyquist Design, State Space & Microcontroller-Based Motion Control

## LEADERSHIP

**President (previously VP & Director of Finance) | GT IEEE** *May 2026 - Present*

- Managed $50K+ budget, raised $20K+ in sponsorships (Amazon, TI, RTX, 3M), grew RoboTech to 200+ participants.